# HASSAN SAMIR

*Backend Developer (.NET / ASP.NET Core)*

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## SUMMARY

Computer Science undergraduate focused on backend development with ASP.NET Core and Entity Framework Core. Experienced in building secure, high-performance RESTful APIs using Clean Architecture, CQRS, caching, background jobs, and containerization with Docker. Comfortable with unit testing and writing clean, maintainable code. Looking to grow within a professional development team.

## EDUCATION

**Bachelor of Computer Science**

*Faculty of Science, Mathematics and Computer Science Department — Menofia University*

## PROJECTS

**Employee Management System** | *ASP.NET Core MVC & Python (FastAPI)*

- Built a full-featured HR system with a **Face Recognition microservice** (Python/FastAPI) for employee identification and automated attendance tracking.
- Developed an Attendance & Check-In/Check-Out sub-system with a Kiosk interface for streamlined operations.
- Implemented modules for employees, departments, leave requests, and payroll, plus a dashboard for monitoring system activity.
- Secured the system with ASP.NET Core Identity and role-based access control, backed by EF Core and SQL Server.

**Survey (Polls) Management System** | *.NET Web API*

- Built RESTful APIs for creating polls, submitting votes, and tracking results, with JWT/Identity authentication and role-based authorization (Admin/User).
- Implemented **HybridCache** (in-memory + distributed) for poll and results queries, cutting repeated database round-trips and significantly reducing response time on read-heavy endpoints.
- Added **rate limiting** to protect public and authentication endpoints from abuse and brute-force attempts.
- Implemented **global exception handling** (centralized middleware) returning consistent error responses without exposing internal details, improving API security.
- Moved notifications and email sending to **background jobs**, so requests return immediately instead of waiting on email delivery.
- Added **Health Checks** for monitoring the API and database, and **API versioning** to maintain backward compatibility.
- Applied the Result Pattern, pagination, and EF Core with SQL Server for clean, consistent responses.

**Inventory Management System** | *.NET Web API*

- Built RESTful APIs for warehouse operations: product management, stock tracking, and inventory movements (in/out), with role-based authorization (Admin/Manager/User).
- Applied **HybridCache** for frequently-read data (products, stock levels) to reduce database load and improve response time.
- Implemented **rate limiting**, **global exception handling**, **Health Checks**, and **API versioning**.
- Built a low-stock notification system using **background jobs** to send alerts and emails asynchronously without blocking requests.
- Used CQRS, repository pattern, and unit of work for a maintainable architecture; wrote **unit tests** for core business logic and containerized the app with **Docker**.

**Blog Platform API** | *Clean Architecture & CQRS, .NET Web API*

- Designed a scalable Blog API using Clean Architecture and CQRS to separate read/write operations, with endpoints for posts, comments, and users.
- Built a modular structure (Domain, Application, Infrastructure, API) with clear separation of concerns.
- Applied FluentValidation for input validation and Mapster for efficient object mapping between layers.
- Implemented JWT authentication/authorization with consistent error handling and API response patterns.

## COURSES

- Advanced ASP.NET Web API Course (El Helaly) — building scalable backend systems (Survey Basket Management).

## TECHNICAL SKILLS

**Programming Languages:** C#, Python
**Frameworks & Libraries:** ASP.NET Core Web API, ASP.NET Core MVC, EF Core, LINQ, ASP.NET Core Identity, FastAPI, SignalR, FluentValidation, Mapster
**Databases:** SQL Server
**Architecture & Patterns:** Clean Architecture, CQRS, Repository & Unit of Work, Result Pattern
**Performance & Security:** HybridCache, Rate Limiting, Global Exception Handling, JWT, Background Jobs
**DevOps & Testing:** Docker, Unit Testing, Health Checks, API Versioning
**Other:** Data Structures, Algorithms, HTML, CSS, Face Recognition / AI Integration

## LANGUAGES

**Arabic:** Native • **English:** Very Good